

```

function pc = mobMap_denseMatching_helper( disp, K, img, zlimit, B )

N=1;
[v,u] = size(disp);

RGB(1,:) = uint8([0,0,0]);
XYZ(1,:) = double([0,0,0]);
f = K(1,1);
cx = u/2;
cy = v/2;

% cameraMatrix: cam2pix
M = [f, 0, cx; ...
     0, f, cy; ...
     0, 0, 1];

% inverse cameraMatrix: pix2cam
iM = inv(M);

for i = 1 : 2 : u
    for j = 1: 2: v

        d = disp(j,i);
        d_ = double(d);

        % pixel ==> i2c
        p = iM*[i,j,1]';
        % Disparity ==> i2c
        d_camKos = double(d) / f;
        % Formel nach Skript ==> f = 1
        Z = B * 1 / d_camKos * 100; % [cm]

        if ( Z <= 0 || Z > zlimit)
            ;
        end
    end
end

```

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        else
            X = B * p(1) / d_camKos * 100; % [cm];
            Y = B * p(2) / d_camKos * 100; % [cm];
            XYZ_i = single([X,Y,Z]);
            XYZ(N,1:3) = XYZ_i;
            RGB(N,1:3) = uint8(img(j,i, 1:3));
        end
        N = N+1;
    end
end

pc = pointCloud(XYZ, 'Color', RGB);
pc = pcdenoise(pc, 'NumNeighbors', 3, 'Threshold', 0.3);
pc = pcdenoise(pc, 'NumNeighbors', 3, 'Threshold', 0.3);

end

```